Key Function of Respiratory System Is Gas Exchange

- Provides oxygen
- Removes carbon dioxide

Upper Airway - Warms, Humidify, and Filter Air

- Nose
- Mouth
- Pharynx
- Larynx
- Trachea

Lower Airway - Gas Exchange

- Bronchi
- Bronchioles
- Alveolar Ducts
- Alveoli - Functional Unit

Pleura - Surrounds and cushions each lung
Pleura Cavity - The space between the two layers of the pleura

Deoxygenated Blood → (R) Atrium → (R) Ventricle

Lungs → Oxygenated Blood → (L) Atrium

* When the blood passes to the lungs it passes along the alveoli, this is where diffusion takes place into the capillaries and then carbon dioxide moves the opposite direction into the alveoli to breath out the carbon dioxide

Ventalation: Flow of air in and out of the alveoli

- Asthma causes bronchoconstriction → Impaired Ventalation

Diffusion: Exchange of O_2 and CO_2 between the alveoli and the RBC in the bloodstream

↳ Pulmonary fibrosis makes diffusion less effective because of the scarring and thickening of the lung tissue

Perfusion: the exchange of O_2 and CO_2 between the RBC and the body's tissue

↳ PAD causes impaired blood flow to those extremities

Diagnostic Tests

- ABGs: Acid Base Balance is tested using a sample of their arterial blood.
  - Tests the oxygenation of critical-care patients and patients w/ acute/chronic respiratory blood
    - pH: 7.35 - 7.45
    - $PaCo_2$: 35-45
    - HCO_3: 22-26
    - $Pa O_2$: 80-100
    - $Sa O_2$: 95-100

- Pulse Oximetry
  - Normal O_2 Level: 95-100
  - COPD Patient: Low 90s

- Pulmonary Function Tests (PFTs)
  - Helps diagnose asthma and COPD